

LAB 5 — Apache Pig User Defined Function (UDF)

Complete Ubuntu practical: start Hadoop, create Java Eval UDF, compile it, create the JAR, upload input to HDFS, run the Pig script, and display the final output.

1. Start Hadoop Services

Command:

```
start-dfs.sh
```

Expected output:

```
Starting namenodes on [localhost]\nStarting datanodes
```

Command:

```
start-yarn.sh
```

Expected output:

```
Starting resourcemanager\nStarting nodemanagers
```

Command:

```
jps
```

Expected output:

```
NameNode\nDataNode\nResourceManager\nNodeManager\nJps
```

2. Check Apache Pig

Command:

```
which pig
```

Expected output:

```
/usr/local/pig/bin/pig (example; your installation may differ)
```

Command:

```
pig -version
```

Expected output:

```
Apache Pig version 0.17.0 (example; your installed version may differ)
```

3. Create Lab Folder

Command:

```
cd ~
mkdir -p pig_lab5/com/lab/pig
cd ~/pig_lab5
pwd
```

Expected output:

```
/home/YOUR_USERNAME/pig_lab5
```

4. Create Java UDF

Command:

```
nano com/lab/pig/CleanDataUDF.java
```

Code to paste:

```

package com.lab.pig;

import java.io.IOException;
import org.apache.pig.EvalFunc;
import org.apache.pig.data.Tuple;

public class CleanDataUDF extends EvalFunc<String> {

    @Override
    public String exec(Tuple input) throws IOException {

        if (input == null || input.size() == 0 || input.get(0) == null) {
            return null;
        }

        String str = (String) input.get(0);

        String cleaned = str
            .replaceAll("[^a-zA-Z0-9 ]", "")
            .trim()
            .toUpperCase();

        return cleaned.isEmpty() ? null : cleaned;
    }
}

```

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5. Find Pig JAR

Command:

```
find / -name "pig.jar" 2>/dev/null
```

Expected output:

```
/usr/local/pig/pig.jar (example)
```

Command:

```
PIG_JAR=$(find / -name "pig.jar" 2>/dev/null | head -n 1)
echo "$PIG_JAR"
```

Expected output:

```
/usr/local/pig/pig.jar (example)
```

6. Compile Java UDF

Command:

```
javac -cp "$PIG_JAR" com/lab/pig/CleanDataUDF.java
```

Expected output:

```
No output if compilation succeeds.
```

Command:

```
ls com/lab/pig
```

Expected output:

```
CleanDataUDF.class
CleanDataUDF.java
```

7. Create UDF JAR

Command:

```
jar -cvf CleanDataUDF.jar com/lab/pig/CleanDataUDF.class
```

Expected output:

```
added manifest
adding: com/lab/pig/CleanDataUDF.class
```

Command:

```
ls -lh CleanDataUDF.jar
```

Expected output:

```
CleanDataUDF.jar is listed.
```

8. Create Input Data

Command:

```
mkdir -p input
cat > input/raw_logs.csv <<'EOF'
1,Valid user session active!!!
2,Purchase completed successfully!!!
3,!!!
4,
EOF
```

Command:

```
cat input/raw_logs.csv
```

Expected output:

```
1,Valid user session active!!!
2,Purchase completed successfully!!!
3,!!!
4,
```

9. Put Input into HDFS

Command:

```
hdfs dfs -mkdir -p /user/hadoop/pig_input
```

Command:

```
hdfs dfs -put -f input/raw_logs.csv /user/hadoop/pig_input/
```

Command:

```
hdfs dfs -ls /user/hadoop/pig_input
```

Expected output:

```
raw_logs.csv
```

Command:

```
hdfs dfs -cat /user/hadoop/pig_input/raw_logs.csv
```

Expected output:

```
1,Valid user session active!!!
2,Purchase completed successfully!!!
3,!!!
4,
```

10. Create Pig Script

Command:

```
USERNAME=$(whoami)
echo "$USERNAME"
```

Expected output:

Your Ubuntu username, for example: junaid

Command:

```
cat > filter_script.pig <<EOF
REGISTER '/home/$USERNAME/pig_lab5/CleanDataUDF.jar';

raw_data = LOAD '/user/hadoop/pig_input/raw_logs.csv'
USING PigStorage(',')
AS (id:int, message:chararray);

cleaned_tuples = FOREACH raw_data GENERATE
    id,
    com.lab.pig.CleanDataUDF(message) AS clean_msg;

filtered_data = FILTER cleaned_tuples BY clean_msg IS NOT NULL;

STORE filtered_data INTO '/user/hadoop/pig_output'
USING PigStorage(',');
EOF
```

Command:

```
cat filter_script.pig
```

Expected output:

The Pig script is displayed with your actual Ubuntu username in the REGISTER path.

11. Remove Previous Output

Command:

```
hdfs dfs -rm -r -f /user/hadoop/pig_output
```

Expected output:

Deleted /user/hadoop/pig_output (if it existed). If it did not exist, that is also okay.

12. Run Pig Program

Command:

```
pig -x mapreduce filter_script.pig
```

Expected output:

Pig displays job progress; successful completion should end with Success!

13. Check Output Directory

Command:

```
hdfs dfs -ls /user/hadoop/pig_output
```

Expected output:

```
_SUCCESS
part-m-00000
```

14. Display Final Output

Command:

```
hdfs dfs -cat /user/hadoop/pig_output/part-*
```

FINAL OUTPUT:

```
1,VALID USER SESSION ACTIVE
```

2, PURCHASE COMPLETED SUCCESSFULLY

Final Lab Flow

Java UDF → Compile → Create JAR → Create CSV → HDFS → Pig Script → Run Pig → Display Output

Note: The Pig installation path and version can differ between Ubuntu systems. The commands above detect pig.jar instead of hard-coding a path. Do not type expected-output lines as commands.